GIG-I

Parametric Insurance for the Gig Economy

AI-Powered Income Protection · Adversarial Fraud Defense · Zero-Touch Payouts

Protecting India's 15 Million+ Gig Delivery Workers

Team DevVerse

TEAM DevVerse:

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Backend Developer	Team Leader	Front End Developer	Back End Developer	Tester

THE PROBLEM

India's Gig Workforce Has No Safety Net

Delivery workers earn per-task. When disruptions hit, income stops instantly — but bills don't.

20–30%

Drop in working hours during rain/heat events

₹0

Safety net for gig workers with no traditional insurance

Weeks

Time to settle claims under traditional insurance models

Traditional insurance is too slow, claim-heavy, and not designed for short-term income protection.

OUR SOLUTION

GiG-I— Zero-Touch Parametric Insurance

01
DETECTS

Monitors real-world disruptions via high-fidelity APIs (weather, traffic, govt alerts)

02
PREDICTS

Uses AI (XGBoost + Gradient Boosting) to estimate specific income loss per worker

03
TRIGGERS

Automatically initiates payouts without any manual claim filing by the worker

04
DEFENDS

Prevents pool depletion via 5-layer adversarial fraud detection and graph analytics

"We minimize basis risk by aligning triggers with real income disruption events."

OUR SOLUTION:

1

DETECTS — Proactive Disruption Monitoring:

Traditional insurance is inherently reactive; workers suffer a loss, navigate complex paperwork, and wait for a decision. GiG-I flips this model by being entirely proactive. Our system **DETECTS** disruptions in real-time by continuously monitoring conditions through high-fidelity, third-party APIs. Whether it is an IMD heavy rain alert, a TomTom traffic anomaly indicating flooding, or an official urban shutdown, the system instantly identifies when a specific geographic zone is compromised. We know the disruption is happening before the worker even has a chance to report it.

2

PREDICTS — Precision Income Loss Estimation

Knowing it is raining is not enough; we need to understand the exact financial impact on the individual gig worker. Once an environmental anomaly is detected, our system **PREDICTS** the resulting financial damage. We utilize an XGBoost Classifier for risk prediction and Gradient Boosting Regression to estimate the specific income loss per worker. By analyzing the disruption duration, the worker's historical earning consistency, and their specific vehicle type, the AI calculates a precise, localized Rupee loss estimate rather than relying on broad, inaccurate averages.

OUR SOLUTION:

3

TRIGGERS — The Zero-Touch Payout:

The core promise of GiG-I is eliminating friction. When the predicted income loss hits our parametric thresholds, the system automatically **TRIGGERS** the next phase. It initiates payouts without requiring any manual claim filing by the worker. From the moment the trigger is activated, the backend communicates with the blockchain smart contract to mint the payout token directly to the worker's wallet in less than a minute. They get the funds they need to pay their bills instantly, bypassing the weeks-long settlement times of traditional insurance.

4

DEFENDS — Multi-Layer Fraud Shield

Instant payouts create a massive target for bad actors, so we built GiG-I to aggressively protect its capital pool. The system **DEFENDS** itself using a 5-layer adversarial fraud detection architecture. Before any triggered payout is finalized, it must survive deterministic checks, device integrity scans (looking for emulators or spoofing), and behavioral velocity checks. We also deploy advanced graph analytics to map out IP addresses and device IDs, allowing us to instantly detect and isolate coordinated fraud rings attempting to drain the pool. We block the fraudsters so we can reliably protect the legitimate workers.

END-TO-END WORKFLOW

From Onboarding to Instant Payout — Fully Automated

1 **Onboarding**

2 **AI Risk Profiling**

3 **Weekly Policy**

4 **Real-Time Monitor**

5 **Trigger Activation**

6 **Fraud Validation**

7 **Instant Payout**

END TO END WORKFLOW

•**1. Onboarding:** The gig worker joins the platform, initiating the fully automated workflow from onboarding to instant payout.

•**2. AI Risk Profiling:** An XGBoost model evaluates the worker's location, seasonality, and vehicle type to output a specific Risk Score between 0 and 1.

•**3. Weekly Policy Creation:** The system calculates a dynamic weekly premium policy that is strictly aligned with the gig worker’s payout cycle.

•**4. Real-Time Monitoring:** High-fidelity third-party APIs and edge devices continuously monitor the worker's local conditions and zone for real-world disruptions.

•**5. Trigger Activation:** When a verified disruption occurs, such as flooding or an IMD heavy rain alert, a parametric trigger is automatically activated.

•**6. Fraud Validation:** The claim immediately enters the Multi-Layer Fraud Defense Architecture, where it undergoes event, location, device, behavior, and network checks.

•**7. Instant Payout:** If the Fraud Risk Score (FRS) evaluates to less than 0.25, the system auto-approves the claim and a blockchain smart contract instantly mints a token into the driver's wallet.

PARAMETRIC TRIGGERS

Region-Specific Calibration Aligned with IMD Standards

Heavy Rain

> 60 mm/hour
OR IMD Heavy Rain Alert

Direct drop in delivery volume & safety risk

Flooding

Road Closure / Waterlogging Signals

Physical blockage of delivery routes

Extreme Heat

> 42°C + IMD Heatwave Advisory

Significant drop in worker activity & health risk

Urban Shutdown

Curfew / Official Zone Closure

Access to commercial hubs is physically blocked

⚠️ AQI is NOT a payout trigger — used only as a Risk Scoring feature to prevent basis risk & pool drain.

WEEKLY PREMIUM PRICING MODEL

```
Weekly Premium = (Base Rate × Risk Score) + (Coverage Factor × Coverage Amount)
```

AI Adjustments:

📍 Location Risk

Historical disruption frequency in specific geographic clusters and zones.

🌩️ Weather Patterns

7-day predictive forecasting using OpenWeatherMap + IMD data streams.

📊 Worker Activity

Historical delivery consistency, claim history & behavioral patterns.

Pricing Cycle

- Aligned with weekly payout cycle
- Dynamic adjustment per zone
- Risk Score range: 0.0 – 1.0
- Transparent for workers

AI SYSTEM DESIGN

7.1 Risk Prediction Model

XGBoost Classifier

Input Features:

- Location (lat/lon cluster)
- Seasonality & Month
- Vehicle type
- Historical claim rate
- Zone risk index

Output: Risk Score (0 – 1)

7.2 Income Loss Estimation

Gradient Boosting Regression

Input Features:

- Disruption duration (hours)
- Zone × Event type combo
- Worker's average hourly rate
- Historical income data
- Event severity level

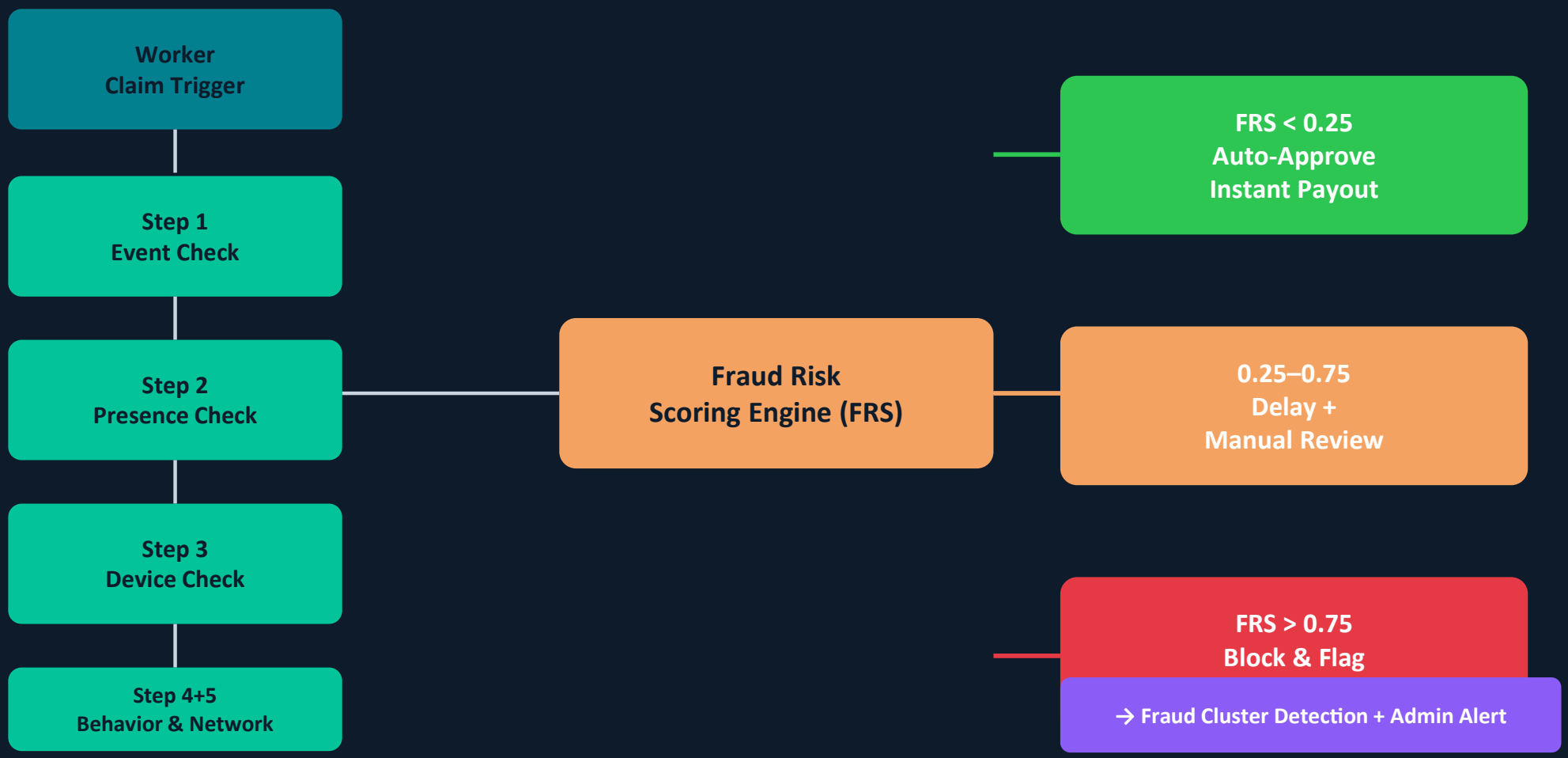
Output: Precise ₹ Loss Estimate

MULTI-LAYER FRAUD DEFENSE ARCHITECTURE

"500 fake GPS claims attempt to drain the payout pool during a legitimate rain event."

Layer 1	**Event Validation**	Signal: Weather / Traffic APIs Detects: Fake or simulated disruption events
Layer 2	**Location Validation**	Signal: GPS Trajectory Analysis Detects: "Teleportation" or perfectly linear movement
Layer 3	**Device Integrity**	Signal: OS / Root Detection Detects: Magisk, Emulators, or GPS Spoofing tools
Layer 4	**Behavioral Model**	Signal: Activity Patterns Detects: "Ghost workers" active only during claim events
Layer 5	**Network Analysis**	Signal: IP / Device Graph Detects: Coordinated fraud rings and Sybil attacks

ZERO-TRUST VALIDATION PIPELINE



FRAUD RISK SCORING (FRS) & DECISION POLICY

$$FRS = w_1(Event) + w_2(Location) + w_3(Device) + w_4(Behavior) + w_5(Network)$$

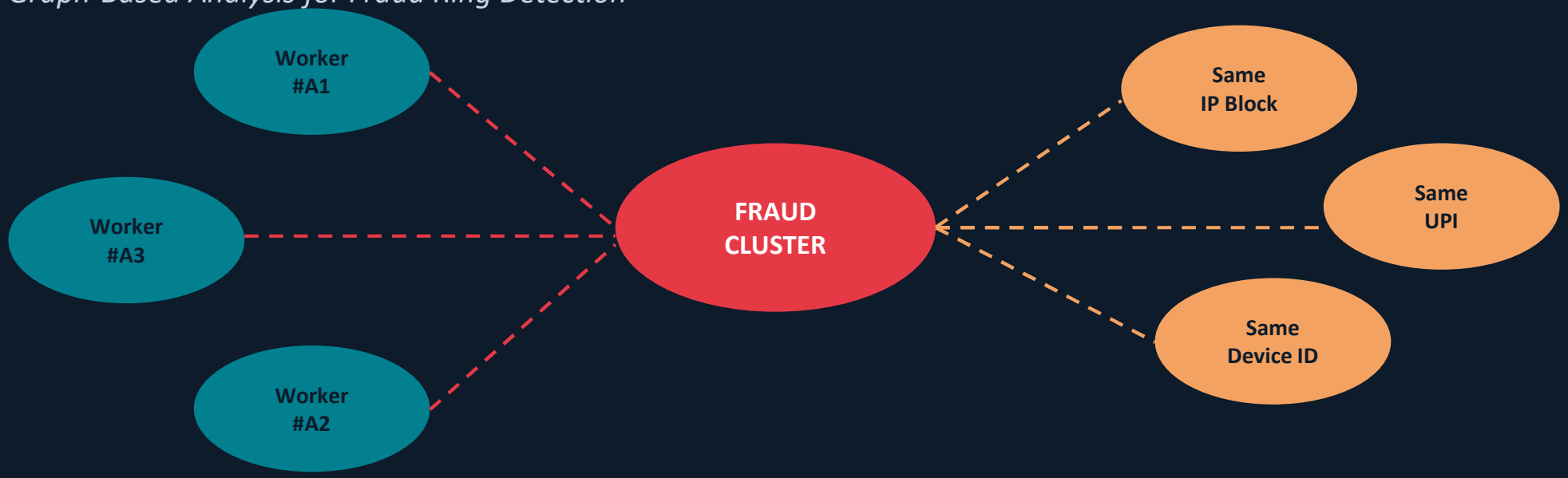
Decision Policy — Research-Calibrated Thresholds

FRS < 0.25	**AUTO-APPROVE**	Instant Payout — No friction for legitimate workers
0.25 ≤ FRS < 0.55	**APPROVE + Monitor**	Silent background monitoring with payout released
0.55 ≤ FRS < 0.75	**DELAYED PAYOUT**	Secondary Validation required before releasing funds
FRS ≥ 0.75	**HOLD**	Pending manual audit — NOT instant rejection (fairness first)

Hard Reject ONLY if FRS > 0.85 AND at least 2 independent signals are strongly anomalous (e.g., Fake GPS + Rooted Device).

COORDINATED ATTACK MITIGATION

Graph-Based Analysis for Fraud Ring Detection



Fraud Ring Signal: Multiple workers + Same IP/Device + Synchronized Claims + Same Zone = Cluster Flagged

- Zone Payout Cap — auto ceiling if claims exceed predicted density by >200%
- Cluster Isolation — instantly disconnects accounts sharing high-risk graph edges
- Manual Override — global switch to shift affected zones to manual verification

FAIRNESS FIRST — FALSE POSITIVE PROTECTION

No single signal can reject a claim — Multi-Signal Independent Validation

GPS Drift Tolerance

±100m allowed for urban 'canyon' signal interference. City infrastructure causes legitimate GPS errors.

Review > Reject

We prioritize investigation over immediate denial. Workers are never silently blocked without cause.

Worker Transparency

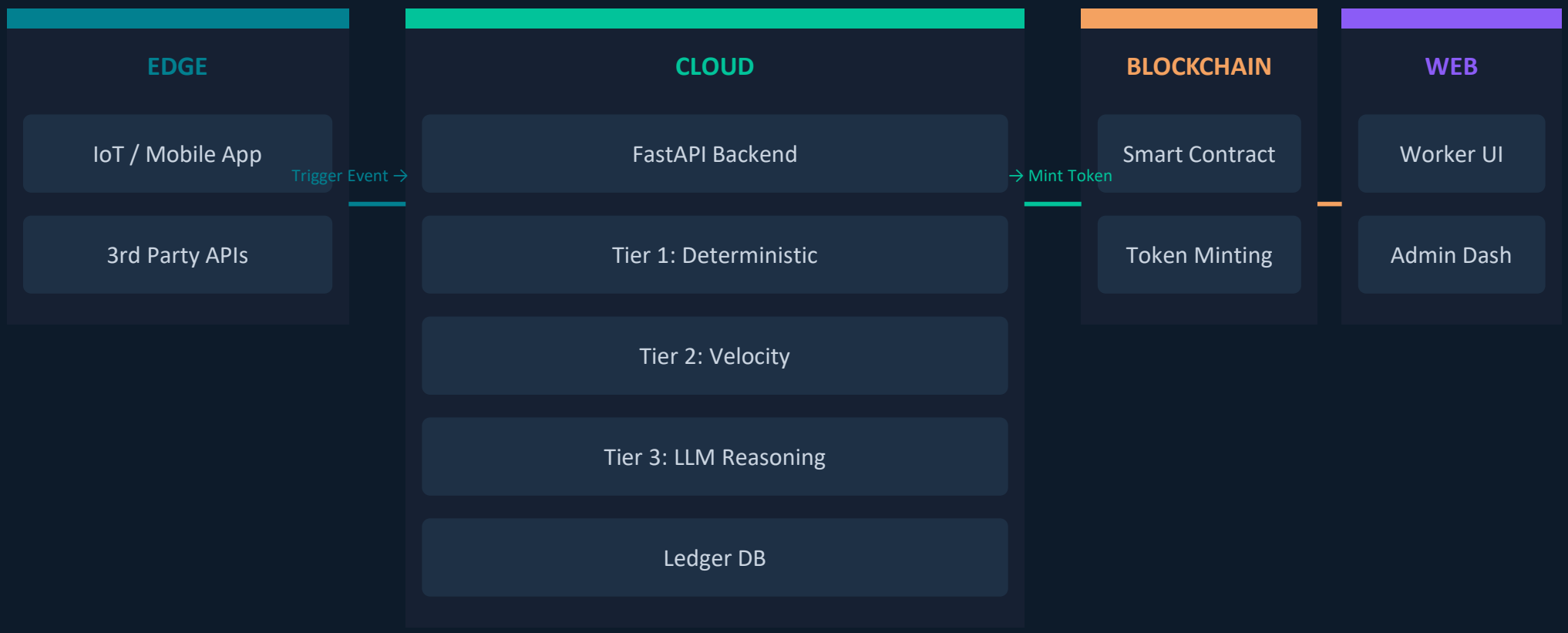
Workers are actively notified of any delays. No silent blocks — full visibility into claim status.

Weighted Signals

Each signal has a calibrated weight. No single anomaly can alone reject a legitimate claim.

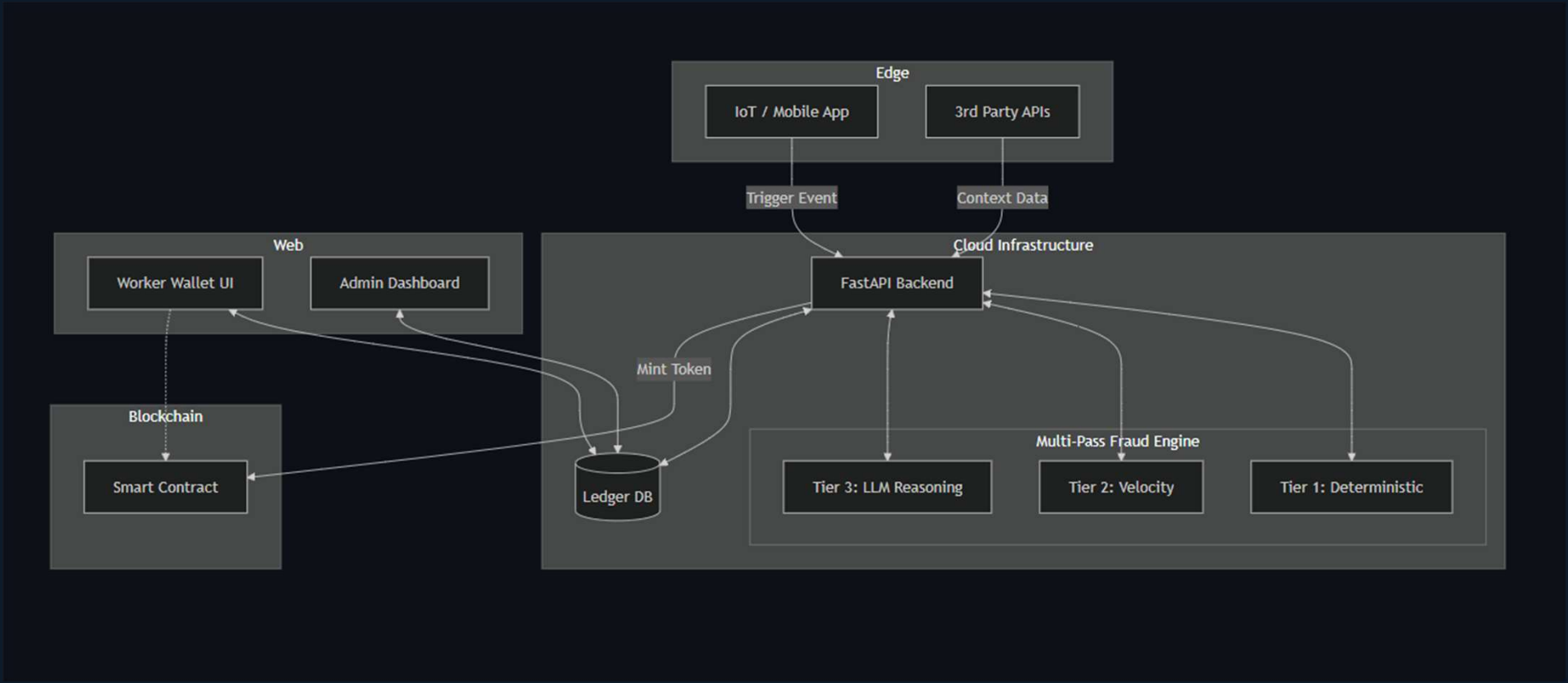
"Our anti-fraud system is designed to protect legitimate workers just as aggressively as it blocks fraudsters."

HIGH-LEVEL SYSTEM ARCHITECTURE



- FastAPI ↔ Tier 1/2/3 Engines
- API → Smart Contract (mint token)
- Dashboard ↔ Ledger DB
- IoT → Backend via MQTT/HTTP

HIGH-LEVEL SYSTEM ARCHITECTURE



GIG-I Architecture Data Flow

The Edge Layer: The workflow begins at the edge, where IoT devices or mobile applications continuously monitor local conditions and send "Trigger Events" to the cloud. This data is enriched by "Context Data" gathered from third-party APIs, which supply real-time information such as weather updates, traffic conditions, and curfew alerts.

Cloud Infrastructure: All incoming trigger events are funneled directly into the FastAPI backend, which is hosted on cloud infrastructure.

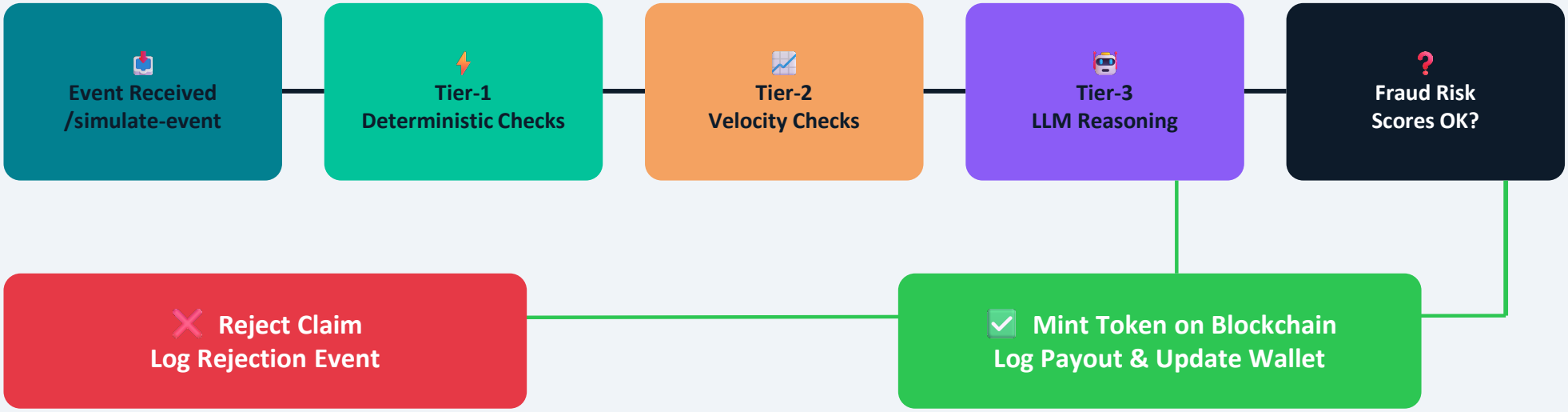
Multi-Pass Fraud Engine: Once an event reaches the backend, it activates the Multi-Pass Fraud Engine. This engine operates as an asynchronous validation pipeline divided into three distinct tiers:

- **Tier 1:** Executes fast, rule-based deterministic checks to verify basic event conditions.
- **Tier 2:** Performs velocity and historical checks by analyzing the driver's recent claim history to prevent rapid-fire exploits.
- **Tier 3:** Acts as the final arbiter by utilizing an LLM Reasoning Agent to conduct a contextual analysis of the event.

Blockchain Integration: If the claim successfully passes all three tiers of the fraud engine, the backend interacts with a blockchain Smart Contract to authorize a "Mint Token" action. This automatically deposits a payout token into the driver's on-chain wallet.

Web Interfaces & Ledger DB: Every payout and token minting event is permanently recorded in the Ledger DB. The web layer, which includes the Worker Wallet UI and the Admin Dashboard, reads directly from this database (or the blockchain) to provide real-time visibility into wallet balances, claim logs, and system metrics.

DATA FLOW — 3-TIER VALIDATION PIPELINE



Tier 1 — Deterministic

Fast rule-based logic. Validates basic conditions: event type, zone match, timestamp validity. Rejects obvious fakes instantly.

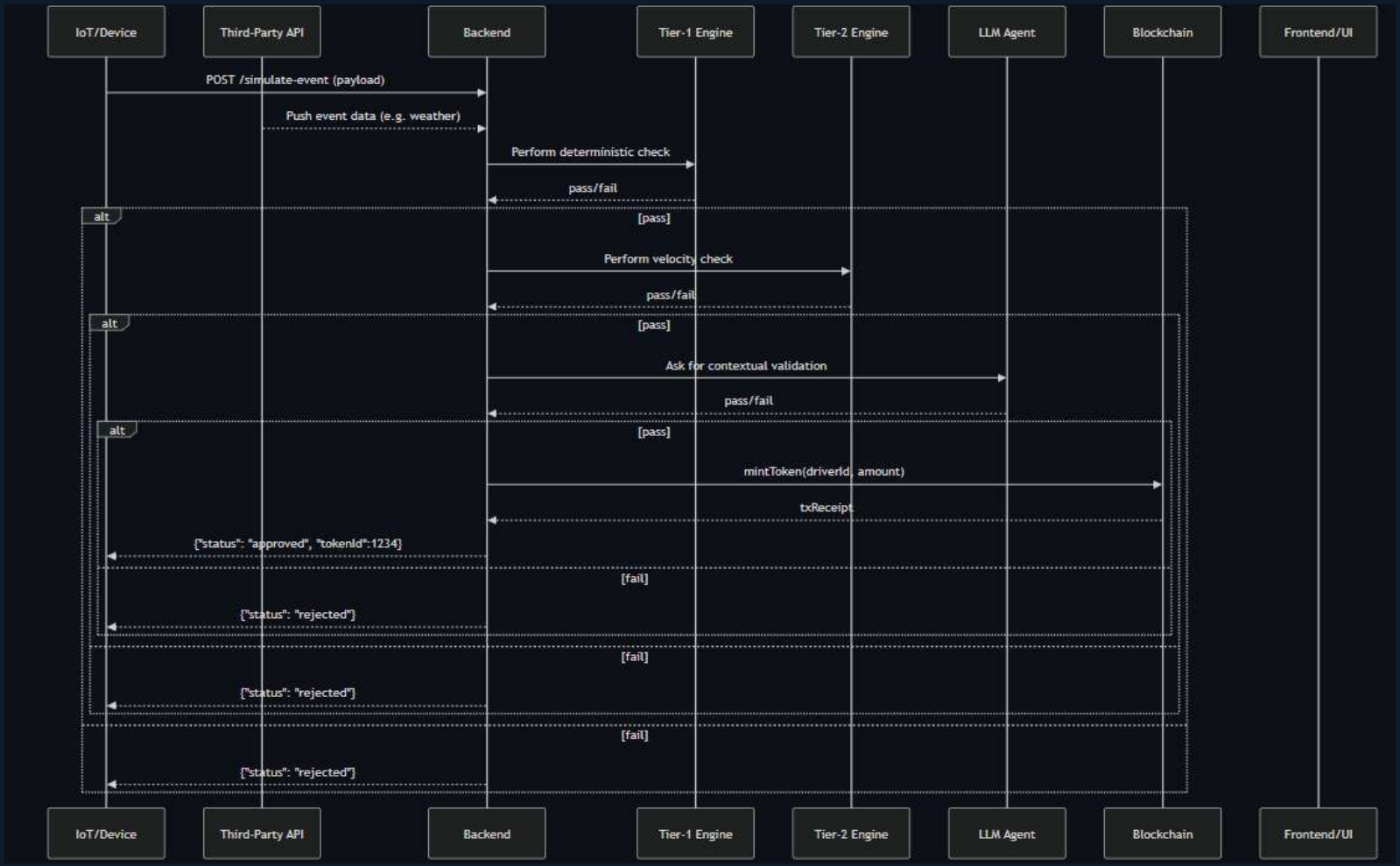
Tier 2 — Velocity Checks

Analyzes driver's recent claim history. Prevents rapid-fire exploit attempts (claim frequency, inter-claim intervals).

Tier 3 — LLM Reasoning Agent

Final arbiter. Structured prompt with event details + external context. Strict I/O sanitization. Only makes yes/no decisions.

END-TO-END SEQUENCE DIAGRAM



END-TO-END SEQUENCE DIAGRAM

- **The 3-Tier Gauntlet:**
  - **Tier 1:** Deterministic logic instantly filters out basic invalid claims.
  - **Tier 2:** Velocity checks block rapid-fire exploits and spam.
  - **Tier 3:** The Nova Reasoning Agent provides final contextual validation

The Trigger:

IoT devices and external APIs (Weather/Traffic) push real-time event data to the backend simultaneously.

Instant Execution:

If all tiers pass, the backend triggers the blockchain smart contract to mint the payout token immediately.

Fail-Fast Architecture:

If any single tier fails, the process short-circuits, instantly logging a rejection and protecting the payout pool.

CYBERSECURITY & DATA INTEGRITY ARCHITECTURE

Data Integrity (SHA-256)

All sensor data & GPS logs hashed with SHA-256 before storage. Prevents post-event tampering of location history.

Secure Communication (TLS 1.3)

All API interactions encrypted via TLS 1.3. Protects against Man-in-the-Middle (MITM) attacks.

IAM — JWT + MFA

Short-lived JWT access/refresh tokens for workers. Admin actions require Multi-Factor Authentication.

Immutable Audit Logging

Full trail for every claim trigger, manual review, and payout authorization. Total accountability.

E2EE — AES-256-GCM

Worker PII and UPI details encrypted at rest with AES-256-GCM. Zero exposure on breach.

Rate Limiting & DDoS Protection

Advanced rate limiting at API Gateway prevents brute-force attacks on the payout trigger engine.

API CONTRACTS & DATA SCHEMAS

POST /simulate-event

Request Fields:

- driverId: integer
- eventType: string
- timestamp: datetime
- location: {lat, lon}
- zone: string (Geofence)

Response:

```
{ "status": "approved", "fraudRiskScore": 0.07, "tokenId": 1234 }
```

GET /wallet/{driverId}

Request Fields:

- Returns: balanceTokens
- transactions[]: tokenId
- type: payout/deduct
- event: HeavyRain etc.
- timestamp: ISO date

Response:

```
{ "driverId": 42, "balanceTokens": 1500, "transactions": [...] }
```

GET /ledger

Request Fields:

- Admin-only endpoint
- entries[]: event
- driverId, time
- tokenId, FRS score
- Full immutable log

Response:

```
{ "entries": [{"event":"Flood","driverId":42,"FRS":0.07},...] }
```

TECH STACK & DEPLOYMENT

Frontend

Next.js — Responsive Mobile-Web App

Backend

Node.js + FastAPI (Python) — Async, Pydantic models

ML Models

Scikit-learn · XGBoost · Prophet (time series)

Database

PostgreSQL (Prisma ORM) · Redis (cache/velocity)

Security

JWT · SHA-256 · TLS 1.3 · AES-256-GCM

APIs

OpenWeatherMap · WAQI (AQI) · TomTom Traffic

Payments

Razorpay (Sandbox Simulation) · UPI integration

Blockchain

EVM-compatible chain · ERC-20/721 Smart Contracts

Cloud-agnostic · Docker/Kubernetes · AWS/GCP/Azure · Region-local data storage · MQTT/WebSockets edge

COMPONENT OVERVIEW & SYSTEM ADVANTAGES

Component	Role / Description
FastAPI Backend	Receives events & orchestrates payout process (async, Pydantic)
Fraud Engine Tier 1	Deterministic business rules and threshold checks
Fraud Engine Tier 2	Velocity/historical checks against database records
Fraud Engine Tier 3	LLM contextual check — secure AI service, vetted data
Smart Contract	Blockchain token minting with tight authorization controls
Database / Cache	PostgreSQL + Redis: policies, claim logs, event data
Web Dashboard	UI for drivers/admins — policy, balance, claim logs

Key System Advantages

- Reduces Basis Risk — Triggers strictly correlated to actual income loss
- India-Optimized — Uses IMD standards and UPI-first payment design
- Network-Level Defense — Detects organized fraud rings, not just individuals
- Zero Friction — Autonomous workflow from trigger to payout in seconds

MISSION & IMPACT

"GIG-I transforms insurance from a reactive claims process into a real-time, AI-driven income protection system — secure against coordinated fraud and optimized for India's gig workforce."

15M+	Zero	5-Layer	< 1 min	100%
Gig Workers In India	Manual Claim Filing Required	Fraud Defense Architecture	Trigger-to-Payout Time	Blockchain Audit Trail